

DARSHAN P KUMAR

SOFTWARE ENGINEER

Padubidri | +91-6360554524 | darshandevadiga2005@gmail.com | linkedin.com/in/darshanpkumar

Summary

Computer Science undergraduate with hands-on experience in Python, Machine Learning, SQL, and REST APIs. Strong foundation in data structures, algorithms, and software development with a passion for building efficient and scalable applications. Eager to contribute to innovative software and AI-driven solutions while continuously learning and growing as a software engineer.

Education

Bachelor of Technology (Computer Science)

NMAM Institute of Technology, Nitte

2023 – Present

CGPA: 8.86 / 10

Pre-University (Science)

Poornaprajna Pre-University, Adamaru

2021 – 2022

Percentage: 89

High School (SSLC)

Sagar Vidya Mandir English Medium School, Padubidri

2010 – 2020

Percentage: 94.4

Experience

Data Science Intern

June 2025 – August 2025

Glowlogics Solutions

- Applied Python, Machine Learning, and Data Analysis techniques on real-world datasets.
- Performed data preprocessing, exploratory data analysis (EDA), feature engineering, and model development.
- Built and evaluated machine learning models using Scikit-learn to improve predictive performance.
- Utilized Python, Pandas, NumPy, Matplotlib, Seaborn, Git, and GitHub following industry-standard development practices.

Projects

PredictPlate – Food Delivery Delay Prediction System | *Python, Machine Learning, Streamlit*

- Developed a machine learning application to predict food delivery time using regression models and delivery status using classification models based on historical delivery data.
- Performed data preprocessing, feature engineering, encoding, and exploratory data analysis (EDA) to improve prediction accuracy.
- Trained and evaluated multiple regression and classification models, selecting the best-performing models using evaluation metrics.
- Built and deployed an interactive Streamlit web application on Streamlit Community Cloud for real-time predictions.
- Tech Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit, GitHub.

VeraDB – Custom Vector Database Engine | *Python, FastAPI, HNSW, NumPy*

- Designed and developed a custom vector database for efficient storage, indexing, and retrieval of high-dimensional vector embeddings.
- Implemented cosine similarity, Euclidean distance, and HNSW indexing to enable fast approximate nearest-neighbor (ANN) search.
- Developed RESTful APIs using FastAPI for vector insertion, search, update, deletion, and metadata management.
- Integrated embedding generation and vector search to support Retrieval-Augmented Generation (RAG) workflows with modular and scalable architecture.
- Tech Stack: Python, FastAPI, NumPy, HNSW, REST APIs, Pydantic, Uvicorn, JSON.

Skills

Programming Languages: Python, C, SQL

Frameworks: FastAPI, Flask, Streamlit

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Databases: MySQL, MongoDB

Tools: Git, GitHub, Postman, Jupyter Notebook

Platforms: Linux/Unix, Windows

Core Concepts: Machine Learning, Data Preprocessing, REST APIs, Data Structures & Algorithms, Vector Databases, HNSW, RAG

Activities & Leadership

Junior JCI President

JCI Club Padubidri

- Led a team to plan and execute community projects, fostering leadership, collaboration, and decision-making skills.

Treasurer

Lions Club Padubidri

- Managed club finances and budgets, ensuring transparency and accountability.

Media Team Member

Association for Computing Machinery (ACM)

- Managed event photography and visual documentation while contributing to promotional activities.

Interests

- Artificial Intelligence, Aviation (AvGeek), Photography, Video Editing

Languages

English, Kannada, Hindi, Tulu